Unit 6, Lesson 4 — API Reference Sheet

Introduction to APIs: Your Programs Meet the Internet

🔑 Key Vocabulary

Term	Definition
API	Application Programming Interface — lets programs request data from servers over the internet
Request	What your program sends to the API ("Give me this data")
Response	What the server sends back (usually JSON data)
Base URL	The server address (e.g., `https://api.exchangerate-api.com`)
Endpoint	The path that specifies what data you want (e.g., `/v4/latest`)
Parameter	Extra info that customizes the request (e.g., `?latitude=40.89`)
Status Code	A number that tells you if the request worked (200 = success)
API Key	A password that identifies your app to the API server
GET	The most common request type — "give me data"
JSON	The format APIs use to send data back (you learned this in Lesson 3!)

🔗 Anatomy of an API URL

```
https://api.exchangerate-api.com  /v4/latest  /USD
└──────────────── base URL ──────────────┘ └ endpoint ┘ └ parameter ┘
```

With Query Parameters

```
https://api.open-meteo.com/v1/forecast ? latitude=40.89 & longitude=-74.04
└──────── base URL ────────┘└ endpoint ┘   └─ parameter ─┘   └─ parameter ─┘
                                          ▲               ▲
                                          ?  starts       &  separates
                                          parameters      parameters
```

📊 Status Codes Quick Reference

Code	Meaning	What Happened
200	✅ OK	Success! Here's your data
201	✅ Created	New data was saved
400	❌ Bad Request	Something wrong with your request
401	🔒 Unauthorized	You need an API key
404	🔍 Not Found	That data doesn't exist
429	🛑 Too Many Requests	Slow down! You're sending too many
500	💥 Server Error	Not your fault — the server crashed

Rule: Always check for `200` before using the data!

🌐 Try These API URLs in Your Browser

Copy and paste each link into your browser's address bar. You'll see raw JSON data — that's the API response!

💱 Exchange Rates (ExchangeRate-API)

What It Does	Click to Try
USD exchange rates	https://api.exchangerate-api.com/v4/latest/USD

What It Does	Click to Try
EUR exchange rates	https://api.exchangerate-api.com/v4/latest/EUR
GBP exchange rates	https://api.exchangerate-api.com/v4/latest/GBP
JPY exchange rates	https://api.exchangerate-api.com/v4/latest/JPY
❌ Fake currency (error)	https://api.exchangerate-api.com/v4/latest/FAKEMONEY

🌍 Country Data (REST Countries)

What It Does	Click to Try
Japan (name, capital, currency)	https://restcountries.com/v3.1/name/japan?fields=name,capital,currencies
Germany	https://restcountries.com/v3.1/name/germany?fields=name,capital,currencies
Brazil	https://restcountries.com/v3.1/name/brazil?fields=name,capital,currencies
India	https://restcountries.com/v3.1/name/india?fields=name,capital,currencies,population
All country info for USA	https://restcountries.com/v3.1/name/united%20states

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🌦️ Weather (Open-Meteo)

What It Does	Click to Try
Hackensack, NJ (°F)	https://api.open-meteo.com/v1/forecast?latitude=40.89&longitude=-74.04&current_weather=true&temperature_unit=fahrenheit
New York City (°F)	https://api.open-meteo.com/v1/forecast?latitude=40.71&longitude=-74.01&current_weather=true&temperature_unit=fahrenheit

What It Does	Click to Try
Tokyo (°C)	https://api.open-meteo.com/v1/forecast?latitude=35.68&longitude=139.69&current_weather=true
London (°C)	https://api.open-meteo.com/v1/forecast?latitude=51.51&longitude=-0.13&current_weather=true

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😂 Programming Jokes (JokeAPI)

What It Does	Click to Try
Random programming joke	https://v2.jokeapi.dev/joke/Programming?type=single
Two-part joke (setup + punchline)	https://v2.jokeapi.dev/joke/Programming?type=twopart
Any category joke	https://v2.jokeapi.dev/joke/Any?type=single

> 💡 **Tip:** Refresh the page to get a different joke each time!

📋 Fake Business Data (JSONPlaceholder)

Good for testing when you don't want to hit a real API:

What It Does	Click to Try
List of users	https://jsonplaceholder.typicode.com/users
Single user (#1)	https://jsonplaceholder.typicode.com/users/1
List of posts	https://jsonplaceholder.typicode.com/posts
Posts by user #1	https://jsonplaceholder.typicode.com/posts?userId=1

📋 JSON ↔ Python Cheat Sheet (Lesson 3 Review)

JSON	Python
`{ }` object	`dict` dictionary
`[ ]` array	`list`
`"string"`	`str`
`123` / `3.14`	`int` / `float`
`true` / `false`	`True` / `False`
`null`	`None`

Parsing JSON in Python

```
import json

# JSON string → Python dictionary
data = json.loads('{"base": "USD", "rates": {"EUR": 0.96}}')

# Access the data
print(data["base"])            # USD
print(data["rates"]["EUR"])    # 0.96
```

🔮 Coming Up in Lesson 5

In the next lesson, you'll use the `requests` library to make these same API calls **from Python**:

```
import requests

# This replaces typing URLs in your browser!
response = requests.get("https://api.exchangerate-api.com/v4/latest/USD")

if response.status_code == 200:       # Check: did it work?
    data = response.json()            # Parse JSON automatically!
    print(data["rates"]["EUR"])       # Use the data!
```

🔗 Free API Directory

All of these APIs are **free and require no API key:**

API	URL	Best For
ExchangeRate-API	exchangerate-api.com	Currency conversion
Open-Meteo	open-meteo.com	Weather data
REST Countries	restcountries.com	Country info
JokeAPI	jokeapi.dev	Fun / testing
JSONPlaceholder	jsonplaceholder.typicode.com	Fake data for testing
Dog CEO	dog.ceo/dog-api	Random dog images